

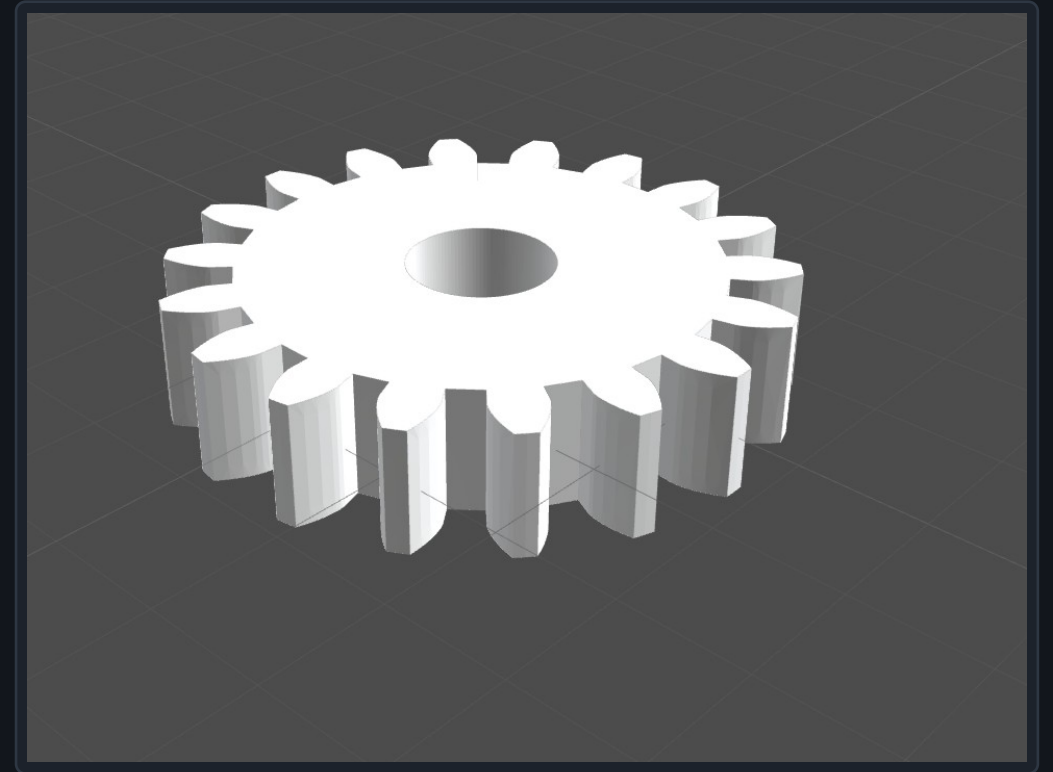
LIVE DEMO

From a sentence to a part you can make

CADSmith turns a plain-language description into CAD geometry, then measures the result in the CAD kernel and rejects it if the dimensions are wrong.

[Presenter name]

[Role · Organisation] · [Event · Date]



SPUR GEAR 50 MM · 2 SECONDS · NO MODEL CALL

THE PROBLEM

A part that looks right is not a part you can make

Generated once, never checked

Most text-to-CAD produces a shape in a single pass. Nothing measures whether it is the shape that was asked for.

A render cannot see millimetres

Visual inspection catches a missing hole. It cannot tell 47.14 mm from 47 mm — and the bolt pattern is the part that has to fit.

Close enough is scrap

In a workshop a plausible part is a wasted billet, a wasted setup and a wasted afternoon.

28.37 mm

mean Chamfer error — one model call, no verification

0.74 mm

the same benchmark with planning, kernel measurement and a vision Judge in the loop

A 38× reduction, from measuring what was built rather than trusting what was generated.

HOW IT WORKS

Five agents and two correction loops

1

Planner

Turns the sentence into a design spec: components, dimensions, constraints.

2

Coder

Writes CadQuery Python, with retrieval over the CAD API documentation.

3

Executor

Runs it in a sandbox, builds the solid in the OpenCASCADE kernel, exports STEP and STL.

4

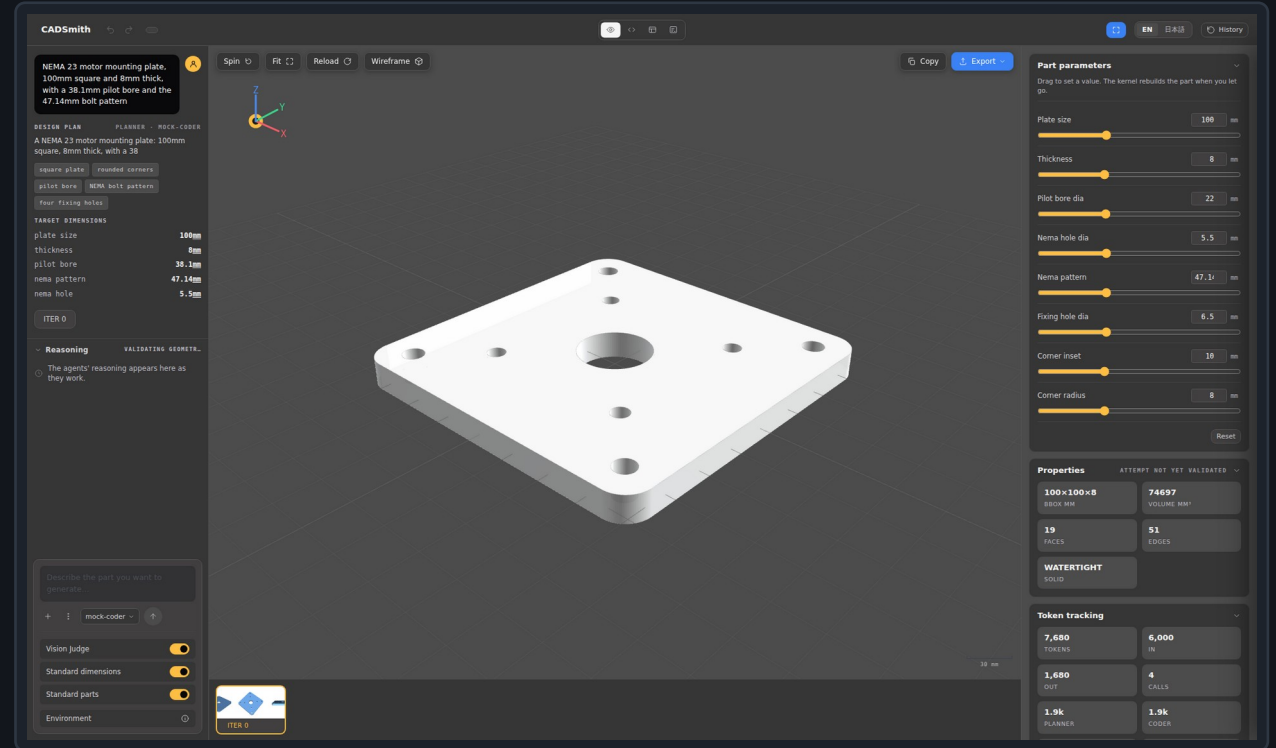
Validator

Measures the solid and shows a vision Judge three rendered views alongside the numbers.

5

Refiner

Gets the exact failure and rewrites the code. Inner loop for errors, outer loop for geometry.



THE PLAN, THE TARGET DIMENSIONS AND THE SPEND, WHILE IT RUNS

Every attempt is kept — including the rejected ones.

DEMO 1 · STANDARD PARTS

Some parts should never be generated at all

An M8 bolt, a 6203 bearing, a 50 mm spur gear: the dimensions come from a published standard, so a model can only introduce error. CADSmith serves these from a catalogue — exactly, instantly, with no model call — and refuses anything ambiguous.

18

part families

2.2 s

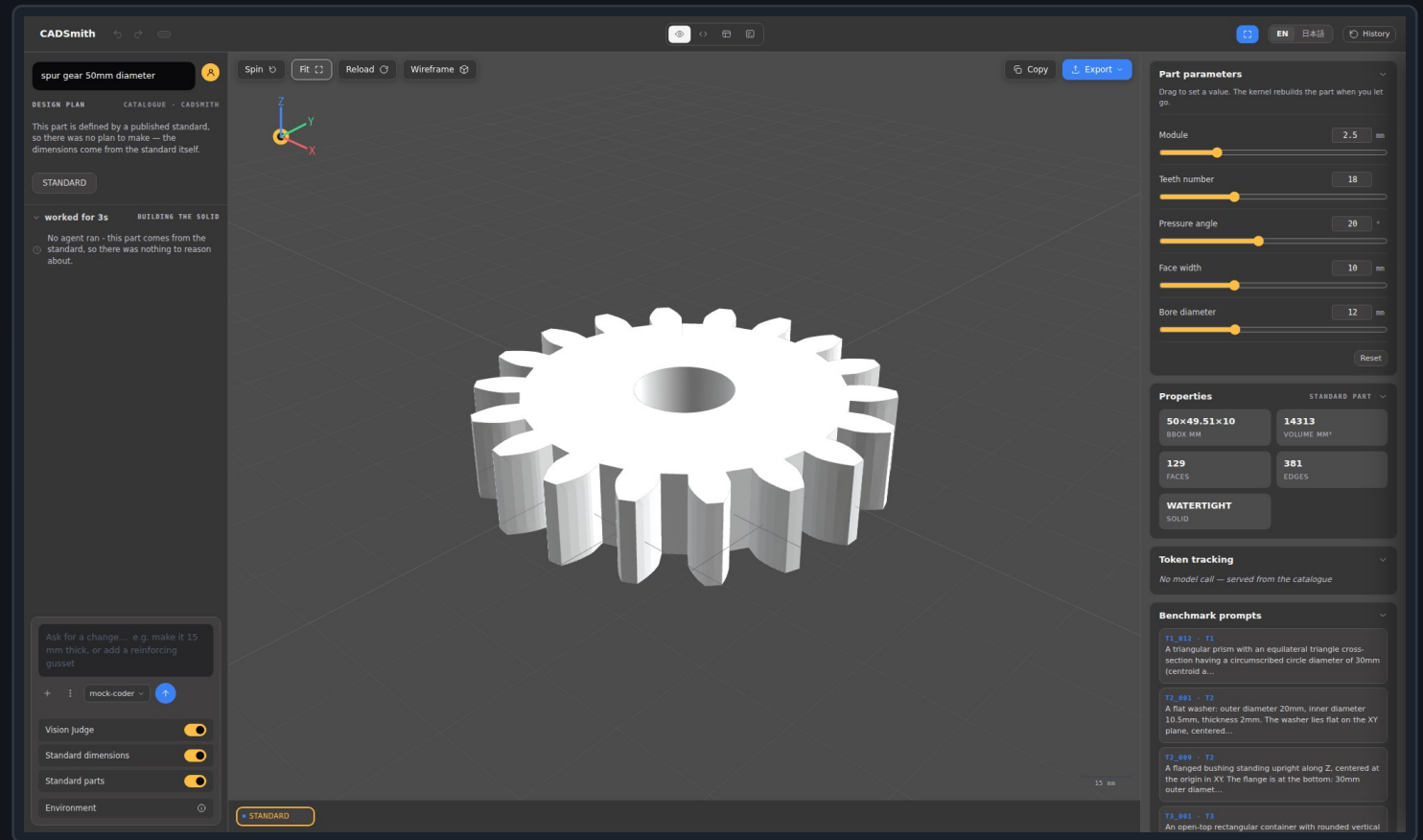
mean time

0

tokens spent

7 / 7

built correctly



BADGED CATALOGUE · NO JUDGE VERDICT · NO MODEL CALL

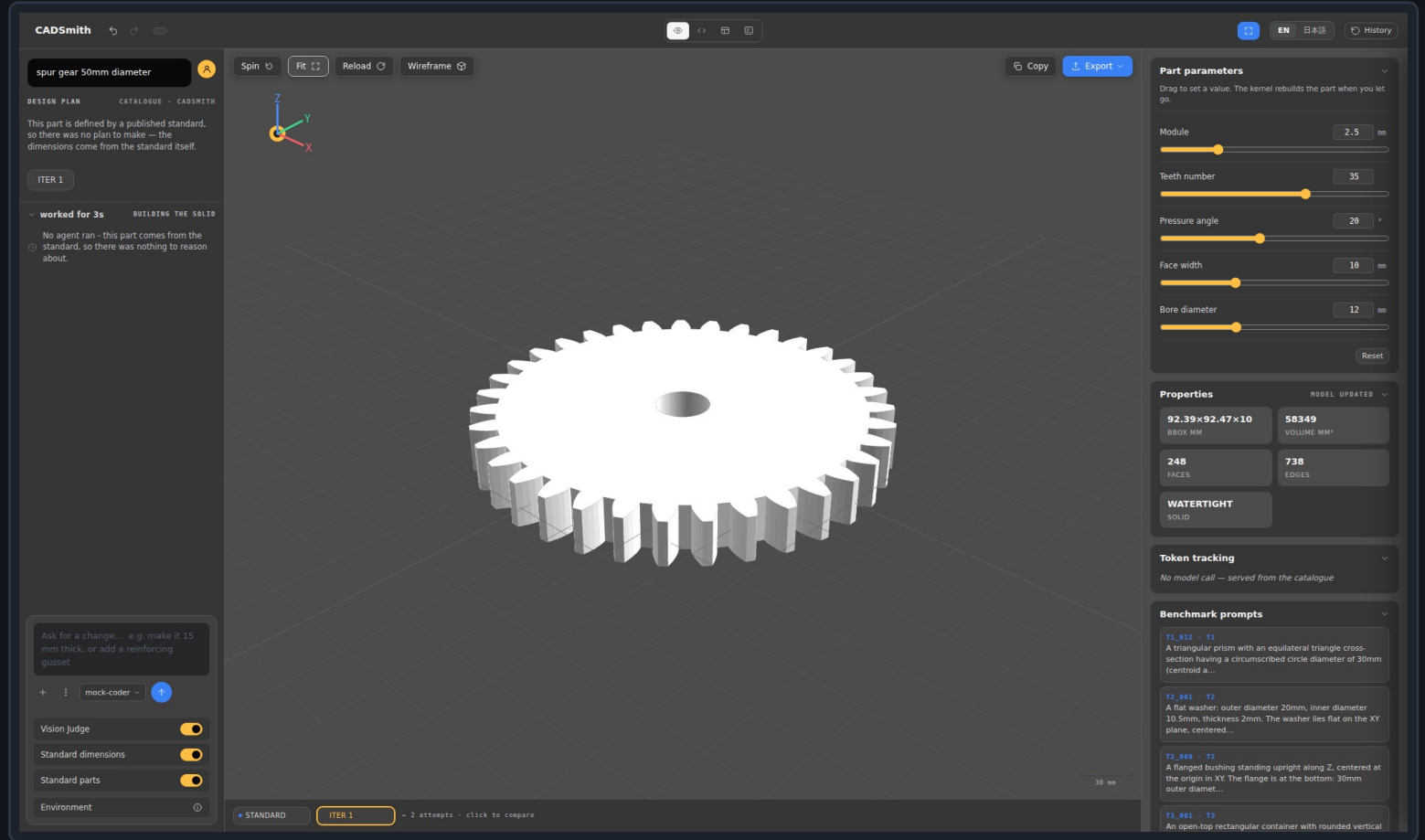
A part no agent produced must never read as evidence that the agents work — so it is badged everywhere it appears, and the catalogue can be switched off to reproduce the published pipeline exactly.

Change a number and the kernel rebuilds the part

The output is source code, not a downloaded mesh. Every named value in the generated script becomes a slider, and letting go rebuilds the solid in about a second — no model call, no new prompt, no lost work.

Here: 18 teeth to 34, and the gear is genuinely re-cut, from 50 mm to 92 mm across.

“Make it 15 mm thick” works too — in English or Japanese — by rewriting the assignment in the script, with no model call at all.



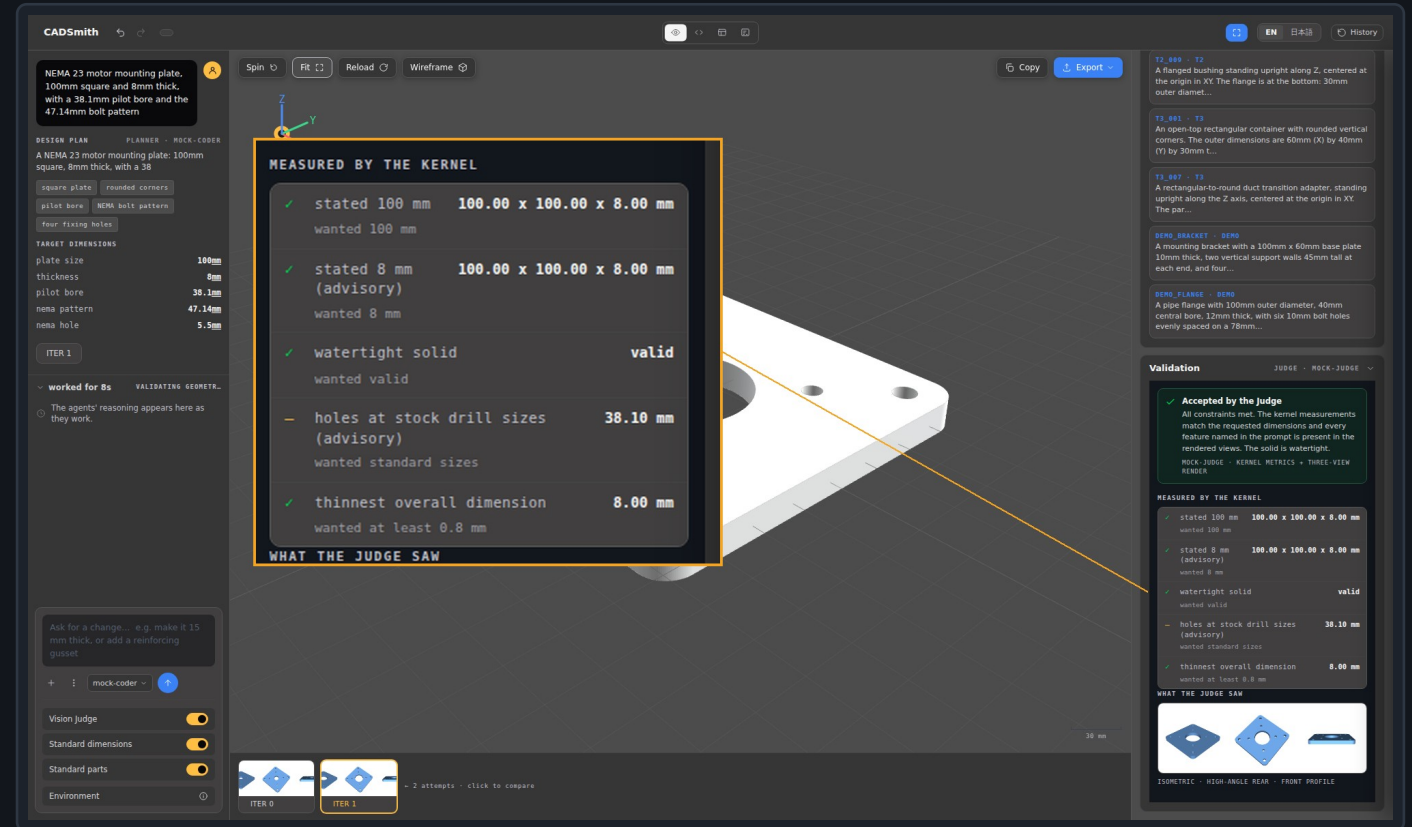
WHAT MAKES IT TRUSTWORTHY

The kernel overrules the Judge

The vision Judge has been seen passing a plate with one hole where four were asked for, and rejecting a part whose volume was exactly right.

So every measurable claim the plan made is measured on the built solid, and the measurement decides. A hole-count shortfall blocks the version however confident the Judge was.

Which checks may block was decided by measurement, not taste. Overall bounding box and volume flagged correct parts in testing, so they are reported and never block — a gate that rejects good work gets switched off, which is worse than having none.



EVERY CLAIM MEASURED · THE JUDGE'S THREE VIEWS · EVERY ATTEMPT KEPT

THE OUTPUT

It ends in a drawing and a file, not a picture

A3 drawing sheet

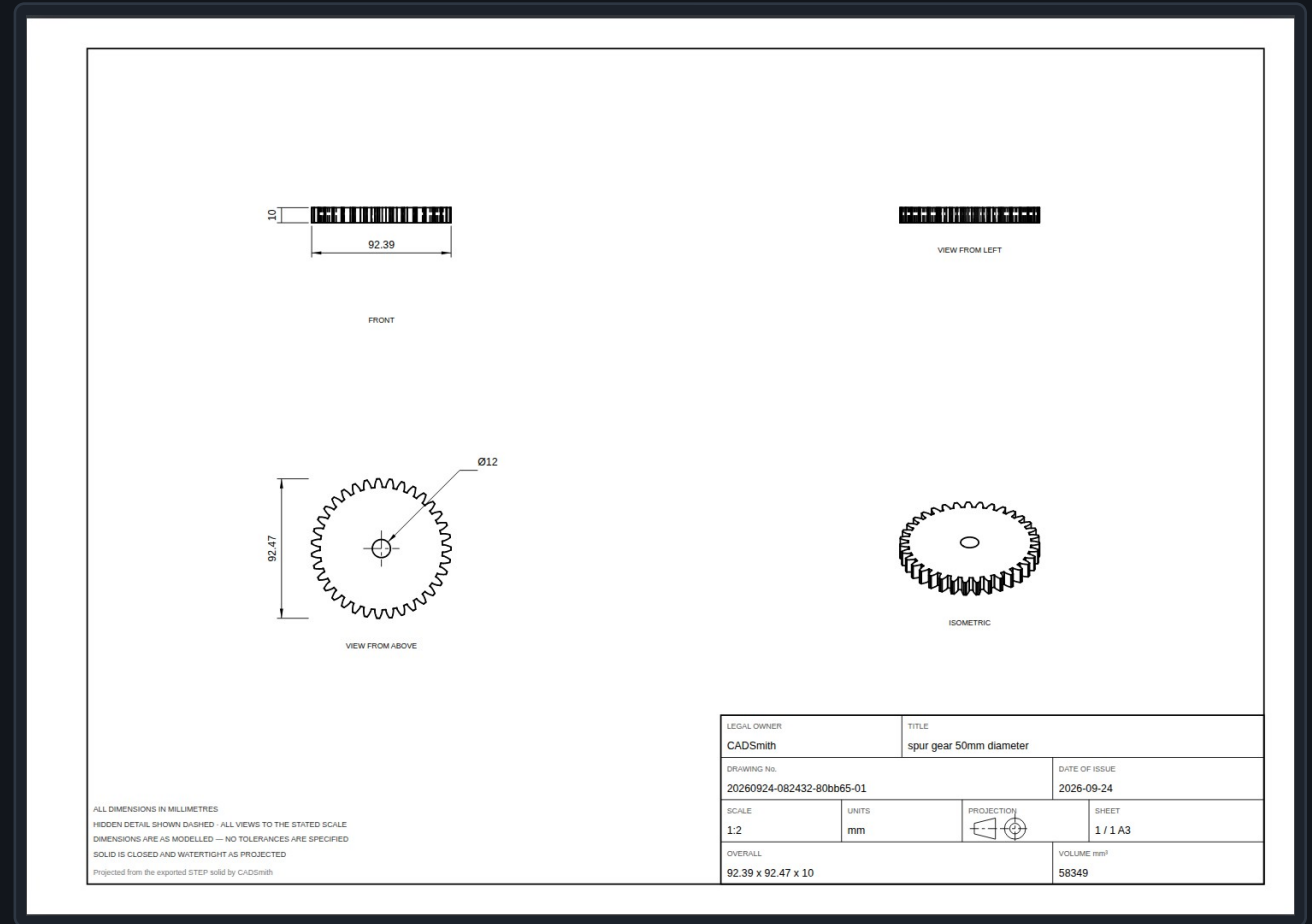
Front, left, top and isometric, projected from the exported solid, with a title block and a first-angle projection symbol. A hole is called out $\varnothing$ and a round R — 4× R5 — each size named once, the way a machinist reads one.

Files a workshop opens

STEP and STL for CAM and printing, DXF for the drawing, GLTF for assemblies — and the CadQuery source itself.

Measurements, not estimates

Bounding box, volume, face and edge counts and watertightness come from the kernel that built the solid.



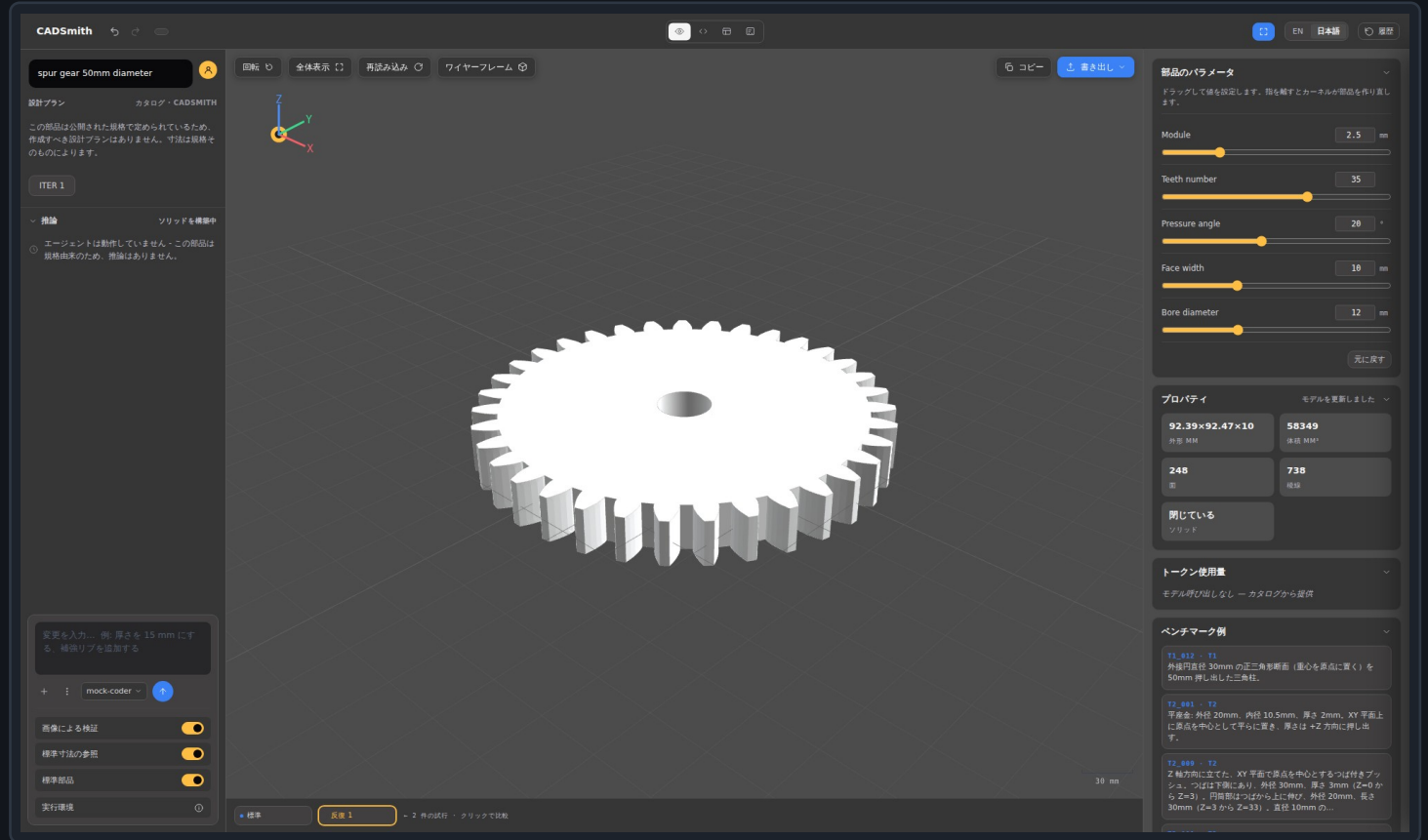
BILINGUAL

English and Japanese, down to the title block

The whole interface, the run log, the verdicts, the kernel facts, the benchmark prompts and the drawing's own title block.

「厚さを 5mm にする」 edits the part without a model call, and 「20 歯 モジュール 2 の平歯車」 reaches the catalogue — while 「20 歯の歯車を入れるギヤボックス」 is correctly read as a gearbox, not a gear.

What is deliberately not translated: what the model reads. The agents are steered in English, and a test fails the build if a Japanese string appears in a module that talks to a model.



SAME PART, SAME NUMBERS, READ IN JAPANESE

The demo still works with the network down

Standard parts

18 families, exact, about two seconds each.

Parameter edits

Sliders and plain-language edits rebuild in the kernel.

Engineering drawings

Projected from any solid already exported.

Exports

STEP, STL, GLTF, DXF and the source.

Recorded runs

Replay a real run — real geometry, real verdicts, marked REPLAY.

Run history

Every past run, every attempt, side by side.

NEMA 23 motor mounting plate,
100mm square and 8mm thick,
with a 38.1mm pilot bore and t...

CONVERGED2VER24/09/2026,
08:38:22▶

NEMA 23 motor mounting plate,
100mm square and 8mm thick,
with a 38.1mm pilot bore and t...

CONVERGED2VER24/09/2026,
08:37:06▶

NEMA 23 motor mounting plate,
100mm square and 8mm thick,
with a 38.1mm pilot bore and t...

CONVERGED2VER24/09/2026,
08:26:41▶

a 60mm square mounting plate,
6mm thick, with a 20mm central
bore and four M5 clearance ...

NOT CONVERGED4VER24/09/2026,
08:25:20▶

spur gear 50mm diameter

CONVERGEDCATALOGUE2VER24/09/2026,
08:24:32▶

a 60mm square mounting plate,
6mm thick, with a 20mm central
bore and four M5 clearance ...

NOT CONVERGED4VER24/09/2026,
08:23:29▶

spur gear 50mm diameter

CONVERGEDCATALOGUE2VER24/09/2026,
08:22:17▶

Every run kept

Converged, not converged, failed or served from the catalogue — in either language, each run keeps its badge, its attempts and its exports, and any of them can be reopened and compared.

Only the five agents need a model backend. Everything a visitor sees in the first sixty seconds needs neither a network nor an API key.

32 prompts, end to end, on one laptop

Tier	Prompts	Built	Correct on every check
Catalogue	7	7	7
Easy	7	7	5
Medium	6	6	1
Hard	3	1	0
Very hard	2	1	1
Total	25	22	14

95 / 112

individual dimensional checks passed

2.2 s

mean time for a catalogue part, 0 tokens

121 s

mean time for an agent-built part (median 110 s)

387,302

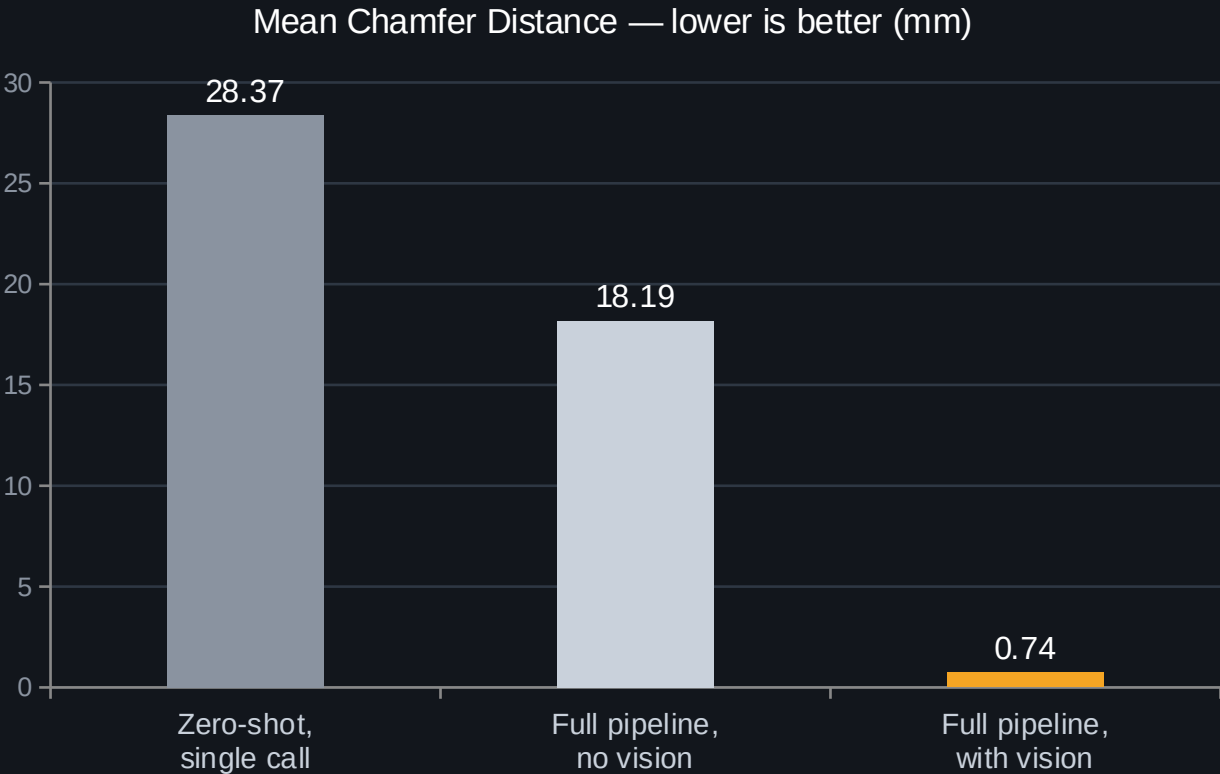
model tokens across 139 calls

Model: Qwen3-VL-8B running on the presenter's own machine — a small open model, not a frontier one. Seven further prompts never reached it, because the tunnel between the app and the model timed out; those are an infrastructure limit of that setup, not a result.

The loop works end to end on a model small enough to run on a laptop. It is not a claim about the state of the art.

THE PUBLISHED RESULTS

What the underlying pipeline scores



Configuration	Executes	Median IoU
Zero-shot, single call	95%	0.8085
Full pipeline, no vision	99%	0.9563
Full pipeline, with vision	100%	0.9629

100 hand-written prompts in three tiers, every reference script written and inspected by hand. Metrics in absolute millimetre space with ICP alignment, so dimensional accuracy counts and not merely shape similarity.

The pipeline is Carnegie Mellon's. The exhibit is the layer around it.

Carnegie Mellon University

Barkley · Loghmani · Barati Farimani — used unmodified

- The five-agent pipeline and its two correction loops
- Retrieval over the CadQuery API for the Coder
- The vision Judge's three-view protocol
- The 100-prompt benchmark and its reference scripts
- The published results

This work

about 22,300 lines of Python and 3,900 of browser code

- The web application: live reasoning, iterations, history, exports
- The standard-parts catalogue — 18 families, and the router that refuses ambiguity
- Measured specification checks that can block a version
- Engineering drawings and DXF
- Parametric sliders and model-free edits
- A per-run token ceiling and per-agent spend
- English and Japanese throughout; Claude on Amazon Bedrock

What it does not do yet

- **Complex parts still fail**

On the hard tiers a small local model builds a solid but misses dimensions. The checks catch it; they do not fix it.

- **No tolerances or GD&T**

Dimensions are as modelled. The drawing says so, in those words, rather than implying a fit it cannot promise.

- **Assemblies are half done**

Clashes are detected and STEP keeps the components separate, but the viewer still shows one fused solid and the drawing has no parts list.

- **Parameter names are the script's**

The sliders are labelled with the identifiers the Coder chose, so they stay English even in Japanese.

- **Speed depends on the backend**

Two seconds from the catalogue; a minute or two for an agent run on a small local model.

A demo that shows its failures is worth more than one that hides them — every rejected attempt stays on screen with the reason it was rejected.

PRESENTER

Who you are talking to

PRESENTER PHOTO

Replace this box with the
headshot supplied by the presenter

[Presenter name]

[Role · Team · Organisation]

Built the exhibited application — the interface, the standard-parts catalogue, the measured checks, the drawings and the bilingual layer — around the published CADSmith pipeline.

[One line on background: what you work on, and why this.]

[email] · [repository or QR destination]

Photo specification portrait or square crop, head and shoulders, 1000 × 1000 px or larger, plain or softly blurred background, even lighting, JPEG or PNG. Landscape crops and phone snaps at an angle look wrong beside the screenshots.

COME AND TRY IT

Type a part. Watch it get made.

1

Say what you need

In English or Japanese, in your own words — no CAD skills required.

2

Watch it get checked

The plan, the code, the kernel's measurements and the Judge's verdict.

3

Take the file with you

STEP, STL, DXF or the drawing — whatever your own tools open.

QR CODE
[repository or contact]

[Presenter name · email]